

# A PCB Test Board for Electrical Characterization of Solder-Bonded Micro-LEDs

Course report — ET4277 + ET4391 — Part 1

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**Abstract**—This work presents a 93 mm × 93 mm two-layer FR-4 PCB designed to characterize how different solder pastes perform when bonding micro-LEDs. The board carries 26 Würth WL-SFCC 0404 RGB LEDs (8 addressable, 18 in two daisy chains), four 0402 NTCs for forward-voltage thermometry, a 100  $\Omega$  0.1% reference resistor with OPEN/SHORT/LOAD pads for impedance spectroscopy, and a set of TLM ladders and van der Pauw cloverleaves on the same substrate. Putting all of this on one PCB means a single cleanroom session yields bond resistance, contact resistivity, sheet resistance, and junction-to-substrate thermal resistance from the same hardware. No separate test coupons, no inter-board correlation. Two 32-pin headers break every active net out to a breadboard. The board is fab-ready for the Aisler or Eurocircuits standard pool with ENIG plating; LEDs are deliberately left unassembled, since the bond joint itself is the research subject and has to form under controlled cleanroom conditions. The report covers the schematic, the characterization methods, the tools needed to run them, and the four-paste comparison the design was built for.

**Index Terms**—micro-LED, die bonding, solder paste,  $V_F$ -TSP, electrochemical impedance spectroscopy, transmission line method, van der Pauw, daisy chain, electrical characterization, PCB test vehicle

## I. INTRODUCTION

Mounting a sub-millimeter LED die on a PCB sounds simple until you measure it. The joint has to carry current with a low voltage drop, pull heat out of the die fast enough to keep it from cooking itself, and survive whatever thermomechanical cycling the application throws at it. With SAC305 reflow paste, the cheap default, the quality of that joint depends on paste rheology, stencil aperture, reflow profile, and substrate finish. Alternative chemistries (Au-Sn eutectic [1], Cu-Cu direct bonding [2], polymer-reinforced pastes [3]) trade one of those parameters for another.

This board extends the v1 design Abdelwahab et al. shipped at ECTC 2025 [4]. The v1 measured bond yield and  $V_F$  only. v2 adds three more characterization methods (TLM contact resistivity, van der Pauw sheet resistance, EIS impedance)

on the same substrate, so one cleanroom session produces a complete electrical fingerprint of whatever paste-and-reflow combination was run. The design deliberately targets the Aisler / Eurocircuits standard pool, not a semiconductor fab, to keep the boards cheap (~€40–50 each) and reproducible at any university lab.

## II. BOARD ARCHITECTURE

The board (Fig. 1) is organized top to bottom into five strips. The top edge has the title block and the north 32-pin header. Of the 32 pins, 8 are pre-routed to the chain endpoints and the NTC sense lines; the other 24 are bare for jumpering, which is useful when bypassing the breadboard to probe a test structure directly. Below the header sits the 6×6 DoE bond-pad array at 3.5 mm pitch, with the EIS calibration block in the same horizontal band: bare OPEN/SHORT pads and a 100  $\Omega$  0.1% reference resistor that doubles as the LOAD reference. The middle strip is the test-structure zone: three TLM ladders at  $W \in \{0.25, 0.5, 1.0\}$  mm with gaps of 200, 300, 500, 1000, 2000, and 4000  $\mu\text{m}$ , plus four VDP cloverleaves at  $W \in \{1.0, 0.5, 0.25, 0.15\}$  mm. The smallest TLM gap stays above the standard-pool minimum at both Aisler and Eurocircuits, so the ladders fab in the same run, on the same ENIG surface, as the LED-side traces. The bottom strip carries the eight WL-SFCC LEDs (D1–D8) with the four NTCs interleaved between them, the DCL6 and DCL12 daisy chains (wired through the  $K_R$  cathode), and the 32-pin south header that pre-routes every D1–D8 signal out to a ribbon cable.

Design rule budget: 0.20 mm minimum copper track, 0.30 mm minimum copper clearance (1.3× and 2× over the 0.15 mm standard-pool minimum at both Aisler and Eurocircuits), 0.30 mm minimum drill, 0.15 mm via annular ring. DRC returns zero violations and zero unconnected items. ENIG plating (Ni 4  $\mu\text{m}$ , Au 0.075  $\mu\text{m}$ ) is non-negotiable: every probe pad has to be gold for reliable probe contact, and the customer-bonded LEDs need a gold surface to wet against. Component count: 130 footprints in total, 7 assembled by the fab (4 NTCs, the reference resistor, and two 32-pin header

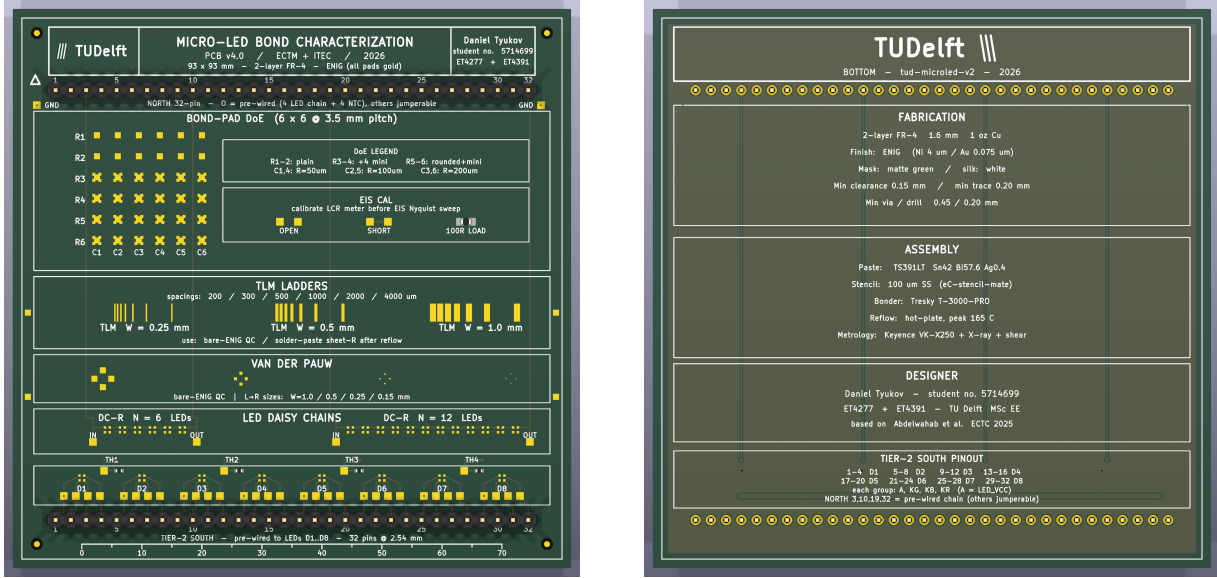


Fig. 1. v4.0.7 PCB, 93 mm  $\times$  93 mm, two-layer FR-4, ENIG finish. Left: top side, with the DoE bond-pad array and EIS calibration block in the upper half, the TLM ladders and VDP cloverleaves in the middle, and the LED row plus NTCs at the bottom. Right: bottom side, with the fabrication / assembly / designer text blocks and the south-header pinout legend that maps the 32 cable wires to the eight LED slots.

strips) and 123 left for the customer (the 26 LEDs bonded in the cleanroom and 97 bare-ENIG test lands).

The companion KiCad schematic captures the 36 electrically active components (the 26 LEDs, 4 NTCs, the resistor, 3 EIS test points, and the two connectors) and the 164 named nets. The remaining 123 footprints are isolated single-pad test structures; they appear on the PCB and in the BOM but adding them to the schematic would clutter it without information, so they are documented through the on-board silk and the project README instead.

### III. CHARACTERIZATION METHODS

#### A. Forward-voltage thermometry ( $V_F$ -TSP)

A diode's forward voltage at a fixed, low sense current falls roughly linearly with junction temperature over a wide range [5]. The slope,  $S \equiv -\partial V_F / \partial T_J$ , is constant enough to invert for  $T_J$  in real time. For a sense current  $I_S \ll I_{op}$ ,

$$T_J(t) = T_{J,0} - \frac{1}{S} [V_F(t) - V_{F,0}], \quad (1)$$

where  $V_{F,0}$  is the forward voltage at the calibration temperature  $T_{J,0}$ . The LED  $V_F$  is read across the four bonded contacts of the device under test via the south header. The substrate reference comes from the four NTCs (TH1-TH4, TDK NTCG104BH103HT1,  $R_{25} = 10$  k $\Omega$ ,  $B_{25/85} = 3380$  K [6]) interleaved between the LED pairs and read through the north header. The Steinhart-B equation maps NTC resistance to temperature:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \ln\left(\frac{R}{R_0}\right), \quad (2)$$

with  $T_0 = 298.15$  K,  $R_0 = 10$  k $\Omega$ . To avoid self-heating bias the sense current is pulsed (100  $\mu$ A, 1 ms, 10% duty),

following the pulsed differential V/I technique of Roscam Abbing and Pertijs [7], which also has the advantage of requiring calibration at only a single temperature. The bond-joint thermal resistance is then the steady-state ratio of the junction-to-substrate temperature difference to the dissipated power:

$$R_{th,bond} = \lim_{t \rightarrow \infty} \frac{T_J(t) - T_{NTC}(t)}{P_{op}}. \quad (3)$$

#### B. Electrochemical impedance spectroscopy (EIS)

EIS sweeps a 10 mV AC excitation from  $10^{-1}$  Hz to  $10^6$  Hz and records the complex impedance  $Z(j\omega) = Z' + jZ''$ . The Nyquist plot  $Z''(Z')$  separates the series bond resistance (the high-frequency intercept,  $Z' \rightarrow R_s$  as  $\omega \rightarrow \infty$ ) from the diode's parallel RC dynamics (the low-frequency semicircle). For the standard LED equivalent circuit ( $R_s$  in series with  $R_p \parallel C_j$ ),

$$Z(j\omega) = R_s + \frac{R_p}{1 + j\omega R_p C_j}. \quad (4)$$

EIS is only as good as its calibration. The board carries three on-PCB references: EIS\_OPEN (two isolated 1.27 mm gold pads,  $Z \rightarrow \infty$ ), EIS\_SHORT (paired pads joined by a solder bridge,  $Z \rightarrow 0$ ), and EIS\_LOAD (the 100  $\Omega$  0.1% reference resistor, Yageo RT0603BRB07100RL, TCR 10 ppm/K [8]). The resistor pads are oversized so they double as four-wire Kelvin probe lands.

#### C. Transmission line model (TLM)

TLM gives sheet resistance  $R_{sh}$  and specific contact resistivity  $\rho_c$  from one set of resistance measurements between

successive finger pairs at known gaps  $d$ . For  $N$  identical contacts of width  $W$  and characteristic transfer length  $L_T$ ,

$$R_{tot}(d) = \frac{R_{sh}}{W} d + 2R_c, \quad R_c = \frac{R_{sh} L_T}{W} \coth\left(\frac{L}{L_T}\right), \quad (5)$$

where  $L$  is the contact length. A linear fit of  $R_{tot}$  vs  $d$  gives  $R_{sh}$  from the slope and  $2R_c$  from the intercept;  $\rho_c$  then follows from  $\rho_c = R_c \cdot W \cdot L_T \cdot \tanh(L/L_T)$ . The three widths in the ladder set make the transfer length's width dependence directly measurable.

#### D. Van der Pauw (VDP)

A symmetric cloverleaf gives  $R_{sh}$  directly, with no contact-geometry correction, from the implicit relation

$$\exp\left(-\pi \frac{R_A}{R_{sh}}\right) + \exp\left(-\pi \frac{R_B}{R_{sh}}\right) = 1, \quad (6)$$

where  $R_A$  and  $R_B$  are transverse four-point resistances measured at  $90^\circ$  rotation. For  $R_A \approx R_B \equiv R$ , this collapses to  $R_{sh} = (\pi/\ln 2) R$ . The four cloverleaves cross-check the TLM slope and surface any width-dependent edge effects.

#### E. Daisy-chain bond yield

DCL6 and DCL12 put 6 and 12 LEDs in series through the  $K_R$  cathode. One I-V sweep across the endpoints (DCL $n$ \_IN/OUT) exercises  $2N$  bonds per chain. A bad bond shows up as a discontinuity in  $I(V)$  or a chain  $V_F$  that overshoots  $N \cdot V_{F,nominal}$ . For WL-SFCC red at 10 mA,  $V_{F,R} \approx 2.05$  V [9], so the DCL12 threshold is roughly 25 V at 10 mA. The  $K_G$  and  $K_B$  cathodes of every chain LED route to isolated probe pads, so a failure can be localized to a specific die and a specific color rather than blamed on the whole chain.

### IV. TOOLING AND WORKFLOW

**Software.** KiCad 9.0.8. The `.kicad_pcb` is written programmatically by a Python script (`tools/generate_pcb_text.py`) rather than placed through the GUI. Fab outputs come out of `kicad-cli`: gerbers and drill (required for Eurocircuits, optional for Aisler since Aisler reads the native `.kicad_pcb`), pick-and-place CSV, top and bottom review PDFs, a 3D STEP model, and the fab-neutral BOM (`tools/gen_fab_bom.py`) which parses identically at Aisler Beagle and Eurocircuits.

**Fabrication.** Either Aisler or Eurocircuits can produce the board on a 2-layer FR-4 1.55 mm pool with ENIG. The fab solders the 4 NTCs, the reference resistor, and the two 32-pin header strips (cut from 1×40 stock). The 26 WL-SFCC LED footprints are marked *do not populate* in the BOM and called out explicitly in the order notes, since the LED bonding is the research subject and has to happen in the TU Delft cleanroom under controlled paste, reflow, and die-bonder conditions. Three boards delivered, fully assembled apart from the LEDs, costs roughly €115–125 at Aisler or €135–160 at Eurocircuits.

**Bonding.** LED placement runs on the Tresky T-3000-PRO. Its  $95 \times 95$  mm work envelope fits the  $93 \times 93$  mm board with 1 mm of margin per side. Solder paste goes down through a  $100 \mu\text{m}$  laser-cut stencil keyed to the WL-SFCC pad layout (four corner pads, each  $0.4 \times 0.4$  mm at 0.8 mm pitch [9]). Reflow follows IPC J-STD-020E with a peak between 245 and 260 °C, the exact profile depending on the paste vendor's datasheet.

**Measurement.** Keithley 2450 SourceMeter for  $V_F$ -TSP and DC I-V. Solartron 1260A frequency response analyzer with a 1296A dielectric interface for EIS. HP4156 parameter analyzer for the TLM and VDP fits. T-type thermocouples soldered to the four TC $n$  gold lands log in-situ reflow temperature, which is useful when a paste lot starts behaving oddly. Probe access is either through 1.27 mm Kelvin tips on the PP\_\* pads or through ribbon-cable breakouts on the 32-pin headers.

### V. SOLDER-PASTE COMPARISON

The motivating experiment is a comparison of four commercial solder pastes on identical hardware: same LED part, same substrate, same reflow profile (within vendor spec), same characterization protocol. The four candidates:

- 1) **SAC305** (Sn 96.5% / Ag 3.0% / Cu 0.5%), the industry baseline.
- 2) **SAC0307**, a low-silver variant. Cheaper, slightly worse on thermal cycling, similar  $V_F$  drop.
- 3) **Au-Sn eutectic**, the canonical micro-LED flip-chip bonding alloy, with reported low post-bond leakage current and high shear strength but a strong sensitivity to bonding temperature [1].
- 4) **Polymer-reinforced solder paste**, an impact-tolerant chemistry developed for laser-assisted mass transfer that preserves solder-joint reliability while absorbing chip-drop energy [3].

For each paste, a three-board batch (1 trial + 1 real + 1 spare, matching Aisler's bundle size and convenient at Eurocircuits too). Per board, measure: chain yield ( $N_{ok}/N$  across both DCL6 and DCL12), per-LED  $V_F$  at 10 mA,  $R_{th,bond}$  from the  $V_F$ -TSP transient,  $R_s$ ,  $R_p$ , and  $C_j$  from a fit of EIS data to (4), and  $\rho_c$  from a fit of TLM data to (5). Cross-board variance is the headline number, since it captures how wide each paste's process window actually is.

The same hardware supports a few other experiments. The  $6 \times 6$  DoE bond-pad array (plain, minis, rounded variants) quantifies capillary self-alignment yield against pad geometry. The four TC pads let the reflow profile be measured and iterated against the paste datasheet. The daisy chains, driven with a programmable current source, run accelerated thermo-mechanical fatigue on a known set of bonds.

### VI. CONCLUSION

This v2 board puts four characterization methods ( $V_F$ -TSP, EIS, TLM, VDP) and a daisy-chain go/no-go test on a single  $93 \times 93$  mm substrate, so one cleanroom session produces a self-consistent measurement of bond resistance, contact resistivity, sheet resistance, and thermal resistance from the

same hardware. The board is fab-ready on the standard pool at both Aisler and Eurocircuits with ENIG plating, with the design rules cleared at twice the published manufacturing minima. The schematic, BOM, and tool chain are committed alongside this report so anyone with KiCad and an account at either fab can re-fab a copy. Part 2 will report the first measurement results from the four-paste comparison once the boards land and the cleanroom session is done.

#### ACKNOWLEDGMENT

This work builds on the v1 board of A. Abdelwahab [4].

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